

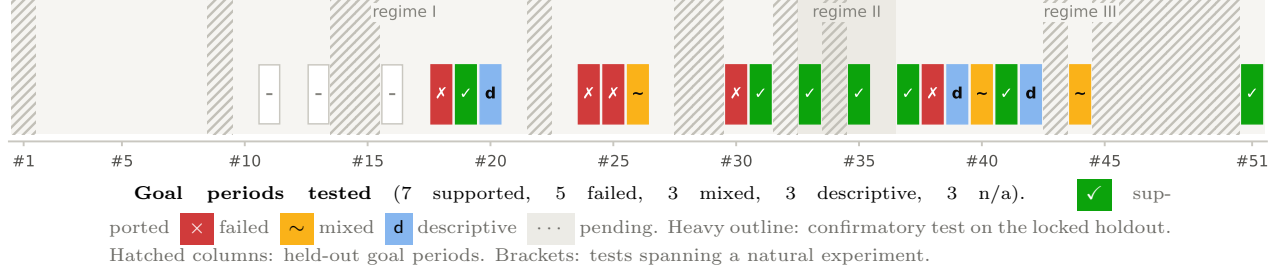
H11: Division of labor vs herding is the sign of a Potts coupling, set by the goal

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. LLM agents herd onto shared artifacts by default, in commits and in attention, and spread out only when the task gives each agent its own.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The spin is the project that an agent touches most in a 30-min window and room (strict artifact mentions). A uniform Curie–Weiss Potts model is fitted per week in 14 non-holdout weeks. The prediction: the sign of βJ follows the goal mode. **It fails:** $\beta J_{CW} > 0$ in 11/14 weeks whatever the mode (mode rule 5/11 signs, $p = 0.73$). Coupling beyond each agent’s own preferences beats circular-shift and day nulls in 10/11 shared-artifact weeks. Spread appears only in own-artifact weeks, as fixed specialization. **Round 1b** (deterministic labels, DQ4 work ledger, DQ6 ballots) changes no verdict. Agents’ *work commits* herd too: in 4/6 shared weeks, onto the repos they attend (same repo in 84% of agent-windows). The claim that attention herds while work stays private therefore fails, except in own-artifact and private-role weeks. On the correct #26 election the runoff jumps from 1/3 to 7/8 in 83 s.

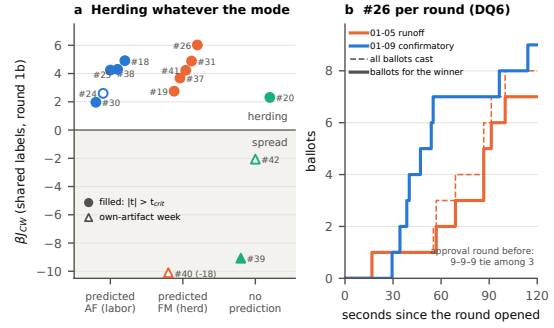
PHYSICAL MODEL

Model 10 (Potts) in Curie–Weiss form. The spin $\sigma_i \in \{0, 1..q\}$ is agent i ’s project in a window ($q \leq 8$; 0 = rare projects, uncoupled). It comes from artifact mentions (attention) or, in round 1b, from agent work commits (work). b is a room block of N_b agents. h_a is the popularity field of project a . βJ is one uniform coupling: > 0 is herding, < 0 division of labor. βJ_c and βJ_s are the mean-field first-order point and the ordered spinodal. A variant with agent fields $h_i(a)$ separates coupling from specialization.

$$P(\sigma) \propto \exp \left[\sum_i h_{\sigma_i} + \frac{\beta J}{N_b - 1} \sum_{i < j \in b} \delta_{\sigma_i \sigma_j} [\sigma_i \geq 1] \right]$$

$$x_a = \text{softmax}_a(h_a + \beta J x_a), \quad x_1 : \frac{1}{q} \rightarrow \frac{q-1}{q} \text{ at } \beta J_c$$

$$\beta J_c = \frac{2(q-1)\ln(q-1)}{q-2}, \quad \beta J_s = \min_s \frac{1}{s} \ln \frac{1+(q-1)s}{1-s}$$



Observables. (a) βJ_{CW} per tested week on the round-1b shared labels, grouped by the class that the goal text predicts. Every predicted division-of-labor week (blue) herds. Negative values occur only in own-artifact weeks (triangles), and a within-agent permutation reproduces them. #40 is clipped (−18). (b) #26 on DQ6 ballots: after a 9–9 approval tie, DeepSeek-V3.2 takes 7 of 8 runoff ballots within 100 s (orange) and 9 of 9 in the 01-09 confirmatory vote (blue).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Herding is the default, and it is real work, not only attention. In shared-artifact weeks agents commit to the same repos in the same half-hour, beyond their stable preferences. Their attention labels point at the repos they commit to (84% of agent-windows). Division of labor appears only when the task gives each agent its own artifact or role. There agents still read each other’s work more than they contribute to it. To get parallel work, assign ownership (a post hoc rule). The coupling-beyond-fields z on artifact links is a cheap monitor for duplicated work: $z \geq 2$ flags herding in commits in 4/6 shared weeks. The Potts machinery adds little: no clean phase transition, no separation of drive and coupling, and the one jump (#26) looks like a ballot cascade.

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RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b)	Verdict
P1 sign of βJ_{CW} follows mode	5/11 signs, $p = 0.73$ (unchanged); same sign in 14/14	mixed by rule; failed
P2 coupling beyond fields by mode	AF weeks $z_{N2} > 0$ in 5/5; #38 failed	failed
P3 first-order consensus	0/4 project jumps; #26 vote jump 1/3→7/8 (correct round), at threshold βJ 3.2 [1.05, 6.0] vs 2.75	failed; jump
P5 held-out gain	8/10 → 10/10	supported
R1b-1 replication	14/14 P1, 12/14 P2 identical	supported
R1b-2a work herds (shared)	$z_{N2} \geq 2$ in 4/6	supported
HH266 work private	work $z < 2$ in 2/5 herding weeks	not supported
#26 field, not cascade	runoff voters saw 2.6 earlier ballots; field favoured Gemini	failed
#40 hub less worked	hub 0.73 of work vs 0.74 of attention	failed

GOAL PERIODS IN DETAIL

G18 × 1b: P1 failed (sig.); P2 failed (sig.) (unchanged)
G19 ✓ 1b: P1 supported; P2 supported (unchanged)
G20 d 1b: descriptive (unchanged)
G24 × 1b: P1 failed; P2 failed (sig.) (unchanged)
G25 × 1b: P1 failed (sig.); P2 failed (sig.) (unchanged)
G26 ~ 1b: mixed (native: jump on the right vote; cascade, n...
G30 × 1b: P1 failed (sig.); P2 failed (sig.) (unchanged)
G31 ✓ 1b: supported (native: work herds)
G33 ✓ 1b: supported (work herds)
G35 ✓ 1b: supported (work herds)
+11 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

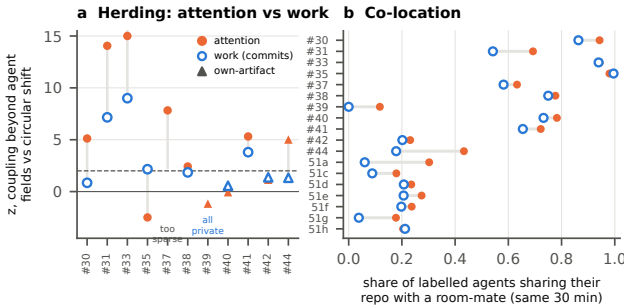
Drive and coupling are still not separable: herding is an effective coupling. z values near 2 move by up to 0.5 between null draws (#20, #38). Work is public agent commits only (dense from #30; author time). In #51 the work-space pseudo-likelihood is ill-determined because each agent has its own repo, so co-location is the measure there. #35, a shared week not used in round 1, has $z_{N2} = -2.5$ in attention. That value would refute the frozen confirmatory clause C2; disclose it before the holdout run. The #26 cascade reading rests on 8 ballots and no text. The ownership rule remains post hoc.

ROUND 2

Fit a kinetic Potts model with ledger read-outs for ballots and commits, to test whether a read of k earlier ballots or commits raises adoption. Use file-level states inside shared repos to look for division of labor below the repo. Use NE15 as an intervention.

- **Essence.** Why do agents pile on: imitation, or attraction to where the work already is?
- **H11-R1** Preferential attachment: $P(\text{join } X) \propto (\text{recent activity on } X)^\alpha$.
- **H11-R2** Herding is stigmergic, not social: artifact activity predicts joining better than chat mentions of X (E3, vs H28).
- **H11-R3** Herding is a coordination solution that raises output; own-artifact spread duplicates effort (test against the work ledger).

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Second figure. (a) Coupling beyond agent fields (z against circular shift) in attention (filled) and work (open), #30–#44. Work herds where attention does, except #30 and #38. Own-artifact weeks (triangles) stay below 2. (b) Co-location: agents that share their repo with a room-mate in the same 30 min. It is equal in shared weeks and lower in work in #44 and in every #51 unit.